

SOFTWARE ENGINEERING

Lab 3

SRN-PES2UG24CS210

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Architecture Selection: We chose Microservices Architecture for the Search Engine System.

Architectural Choice: The search engine is divided into independently deployable services. The main components are the Search UI, Query Processing Service, Search/Ranking Service, Web Crawler Service, and Index & Document Store.

Reason 1 – Independent scaling: Search engines receive many user queries while crawling and indexing workloads may occur at different rates. Microservices allow the Query Processing and Search/Ranking services to be scaled independently from the crawler, so resources can be allocated to the busiest part of the system.

Reason 2 – Fault isolation and independent deployment: A search engine contains distinct functions such as crawling, query processing, ranking, and index storage. Separating these functions into services limits the impact of a failure in one service and allows individual services to be updated or deployed without rebuilding the entire system.

Security Advantage: Each service can have its own authenticated API boundary and access permissions. Public users can access the Search API, while crawler and indexing interfaces can be restricted to trusted internal services. This reduces unnecessary exposure of internal indexing operations.

Performance Benefit: The Search/Ranking Service can retrieve results directly from an optimized index rather than scanning all stored documents for every query. Query-processing and search services can also be scaled horizontally to handle concurrent searches.

Conclusion: Microservices are appropriate because a search engine naturally contains separate, high-volume workloads that benefit from independent scaling, fault isolation, and independent deployment.